

4 (**short**) Consider air (taken as a perfect gas) undergoing compression from ambient conditions, $T_1 = 300$ K and $p_1 = 1$ bar, to a final pressure of $p_2 = 4.5$ bar with negligible changes in kinetic and potential energy.

(a) Assuming the process is reversible and adiabatic, calculate the specific work required on the air. [4]

(b) If the process is adiabatic but not reversible and the specific work done on the air is $w = 200$ kJ kg⁻¹, determine the final temperature and specific entropy increase during this process. [6]